RATIONAL FINANCE

FINANCIAL ECONOMICS – SPRING 2026

RISK AND UNCERTAINTY

- Risk: cases when we have well-defined scenarios of future events and we are able to attach probabilities to them
- Example: toss of a coin, you win €50 if Heads, lose €50 if Tails. This is a "fair game", since the expected value is zero $0.5\times50+0.5\times(-50)=0$

- Uncertainty: we do not know probabilities
- Example: toss of a coin, you win €50 if Heads, lose €50 if Tails. The coin is rigged: the probability of Heads is different from 50% but you don't know the probability and you can't try with a long series of tosses

FINANCE: RISK OR UNCERTAINTY?

- In a situation where all events have a stable probabilistic structure, financial markets are characterized by ex-ante identifiable scenarios to which a probability can be assigned on the basis of data analysis. This requires first of all that the economic system as a whole is stable, but this is a hypothesis that is not always plausible when thinking of wars, financial crises, technological innovation, phases of investor panic...

- In what follows we will assume that financial markets are described by risk and not by uncertainty. It is necessary to remember the limits of this hypothesis, particularly in the new phases, for example the crisis of 2008, the peripheral crisis of 2011-2012 (typical cases of market crashes due to aversion to uncertainty)

THE RISK PREMIUM

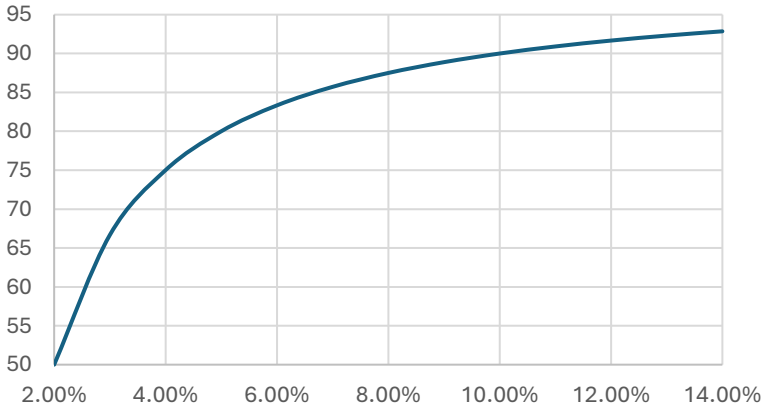
- How much are you willing to pay to participate in fair game? If you are risk-averse, you are not willing to pay anything. Paying e.g. €5, would mean participating in a game that allows you to earn €45 if Heads and lose €55 if Tails, but with an expected value of €-5

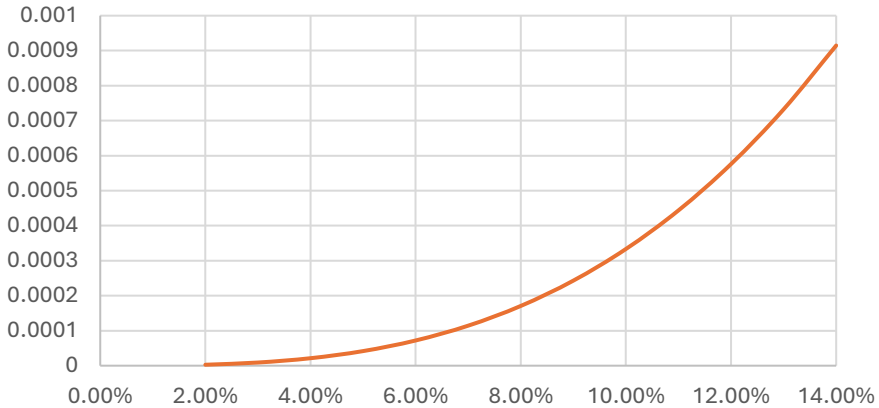
- A risk-averse individual voluntarily participates in a risky game only if the expected value is positive. To voluntarily participate in a risky game when you are risk-averse you need a risk reward

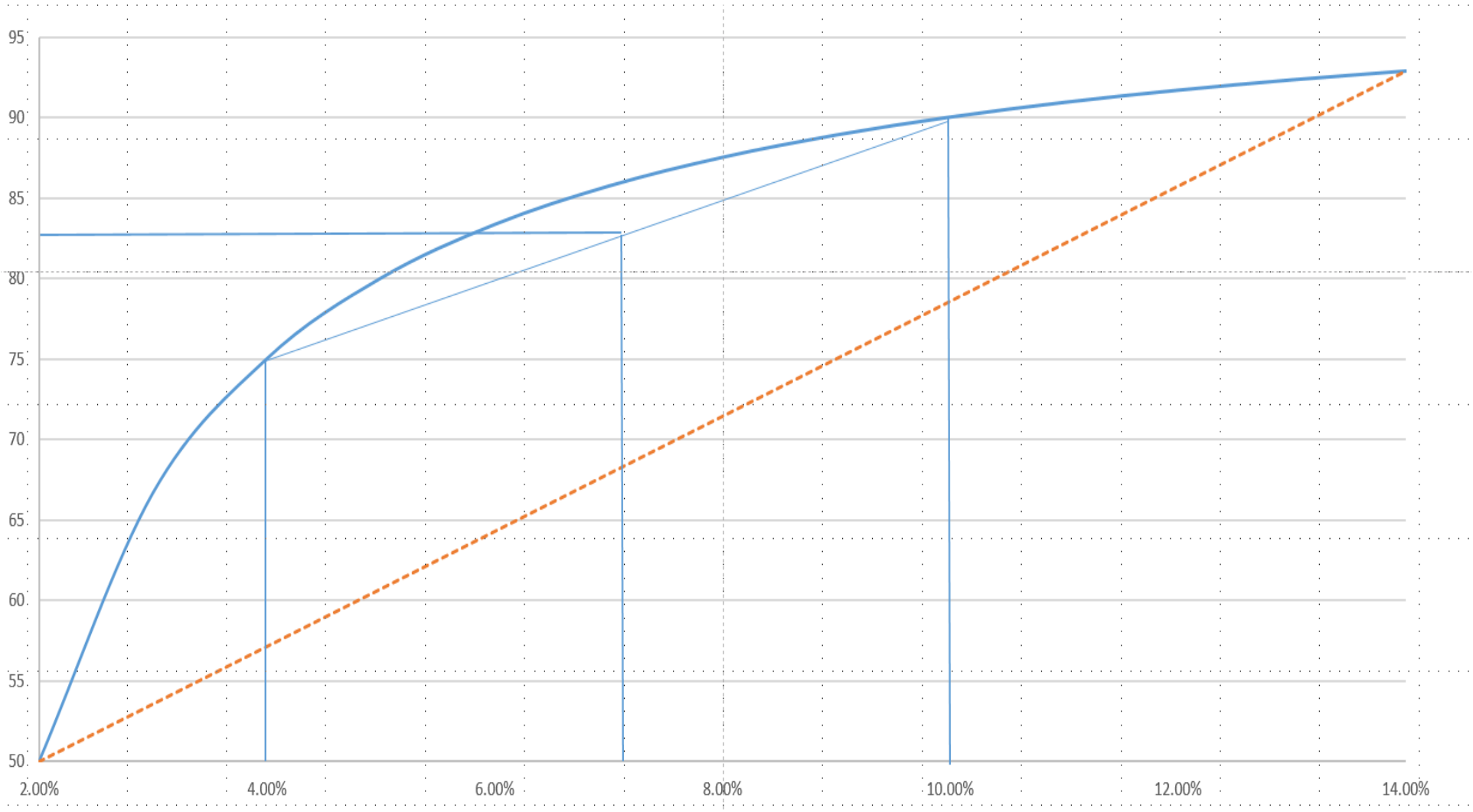
- A risk-averse individual only buys a risky asset if its expected return is higher than the risk-free rate. The risk premium is the difference between the expected return of the asset and the risk-free rate. In gambling, you pay money to take a negative expected return position

THE UTILITY FUNCTION

- The utility function can be defined on the return of the security or portfolio, or on the level of future consumption, i.e. u(C), or future wealth depending on initial wealth and the rate of returnof a risky portfolio $W = W_0 \times (1 + r_p)$
- When the utility function is concave in consumption (or wealth) then we describe risk-aversion. When the utility function is linear, a risk-neutral behavior is described. When the utility function is convex, a risk-friendly behavior is described






95
90
85
80
75
70
65
60
55
50
2.00%
4.00%
6.00%
8.00%
10.00%
12.00%
14.00%

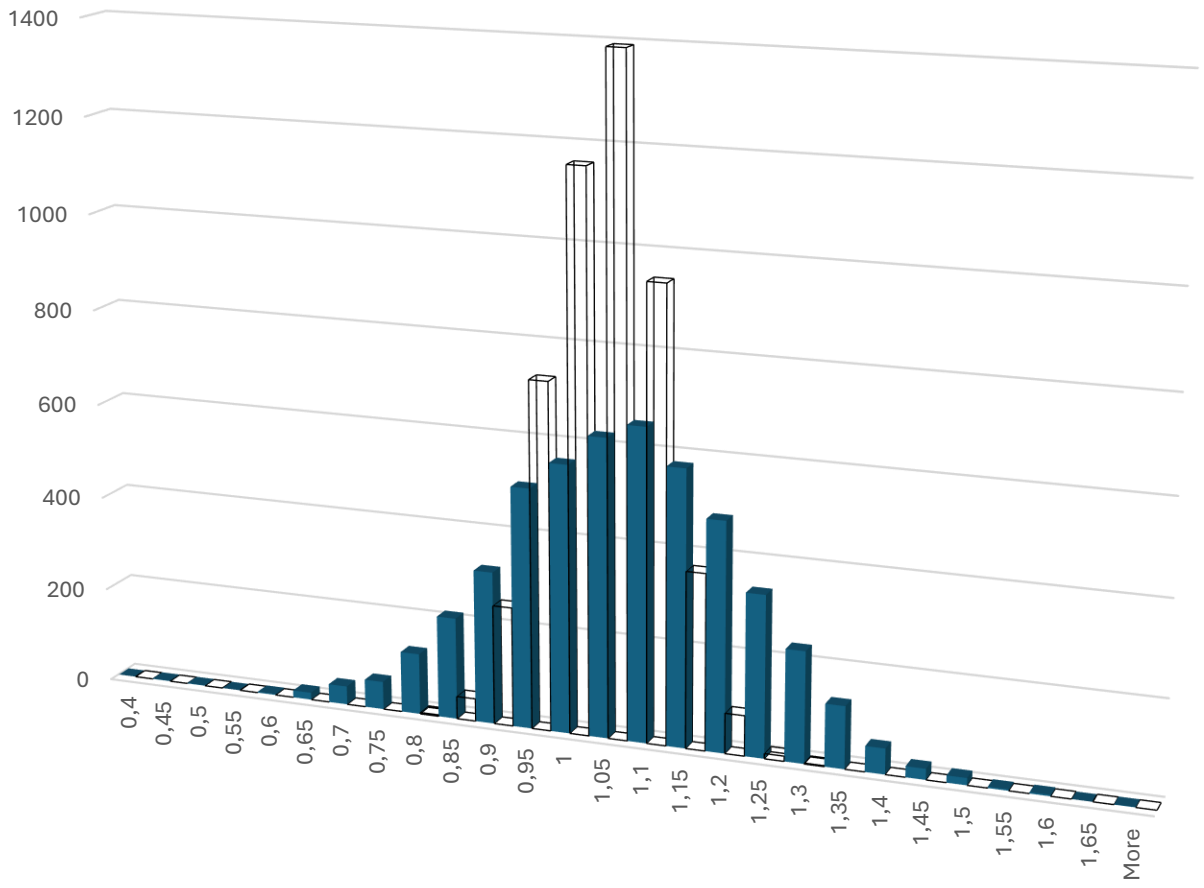
CONCAVE UTILITY FUNCTION

- The utility corresponding to 4% is 75, while the utility corresponding to 10% is 90. If the two events, 4% and 10%, are equally likely, the expected utility is the average of the two levels of utility achieved in each of the two scenarios, and corresponds to the average between 75 and 90, i.e. 82.5.

- But if the individual could have 7% with certainty, the arithmetic mean of 4% and 10%, the utility would be greater than 85. An investor with a concave utility function prefers to have the average value with certainty rather than maintain the risky situation with the associated expected utility.

THE OBJECTIVE

- how do we compare different distributions of final wealth? The optimization criterion is based on maximizing the expected utility Eu(W)



THE CRRA UTILITY

- The constant relative risk aversion (CRRA) utility function has nice properties

- $u(W) = \frac{W^{1-\alpha}}{1-\alpha}$ with $\alpha > 0$ and $\alpha \neq 1$ becoming the log function $u(W) = \ln(W)$ when $\alpha = 1$.

- For this function, the relative risk aversion coefficient $-\frac{W \times u''(W)}{u'(W)}$ is equal to α i.e. i.e. it is independent of the level of wealth, as well as the choice of portfolio. With a CRRA function the choice between a risky and a risk-free security depends only on the risk aversion coefficient and is the same regardless of the level of wealth.

QUESTION

- Three girls have a risk aversion independent of the level of wealth, of the CRRA type. their risk aversion coefficients are 0.5, 3, and 5, respectively, and have to evaluate a risky situation where they can earn €500 or €1000 with equal odds. What is the maximum amount they should they pay to join the game?

ANSWER

- Compute expected utility and make it equal to utility of consumption: for $\alpha \neq 1$

$0.5 \times (1-\alpha)^{-1} 1.000^{(1-\alpha)} + 0.5 \times (1-\alpha)^{-1} 500^{(1-\alpha)} = (1-\alpha)^{-1} C^{(1-\alpha)}$ i.e. solve for $C = (1-\alpha) \times (0.5 \times (1-\alpha)^{-1} 1.000^{(1-\alpha)} + 0.5 \times (1-\alpha)^{-1} 500^{(1-\alpha)})^{1/(1-\alpha)}$

- if α =1 instead $0{,}5 \times \ln(1000) + 0{,}5 \times \ln(500) = \ln(C)$

ANSWER

α	C
0,5	729
3	632
5	586

THE MEAN-VARIANCE UTILITY

- One way to describe the preferences of a risk-averse investor is to consider a utility function that depends positively on the expected return and negatively on the variance of the return (the variance is the square of the standard deviation or volatility). The mean/variance utility function is:

$$U = Er - 0{,}5 \times A \times \sigma^2$$

where Er is the expected return, $\sigma^2$is the variance of the return, A is the risk aversion coefficient, multiplied by 0.5 for the sake of simplicity that will be evident when we calculate the derivative before the portfolio shares when maximizing utility. The higher the value of A, the greater the penalty due to a high level of variance for a given level of expected return

QUESTION

- What investment does an investor with A = 4 choose? What if A=0 (risk neutrality)?

Investment	E(r)	Standard deviation σ	Utility U
1	0.12	0.30	
2	0.15	0.50	
3	0.21	0.16	
4	0.24	0.21	

ANSWER

- What investment does an investor with A = 4 choose? What if A=0 (risk neutrality)?

Investmento	E(r)	Standard deviation σ	Utility U
1	0,12	0,30	-0,0600
2	0,15	0,50	-0,3500
3	0,21	0,16	0,1588
4	0,24	0,21	0,1518

EXAMPLE

- Let's compare, from the perspective of three different levels of risk aversion, an investment in a risk-free security that pays a rate of 2% and a risky investment with an expected return of 10% and a return variance of 2%

Coefficient of risk aversion	Utility risk-free	Utility risky	Preferred
4	2%	$U = 10\% - 2 \times 2\% = 6\%$	Risky
8	2%	$U = 10\% - 4 \times 2\% = 2\%$	Indifferent
16	2%	$U = 10\% - 8 \times 2\% = -6\%$	Risk-free

For an investment in a risky security, the value of the utility function can be interpreted as the certain equivalent, i.e. the value of the interest rate that would make the investor indifferent between buying the risky security and buying the risk-free

INDIFFERENCE CURVE

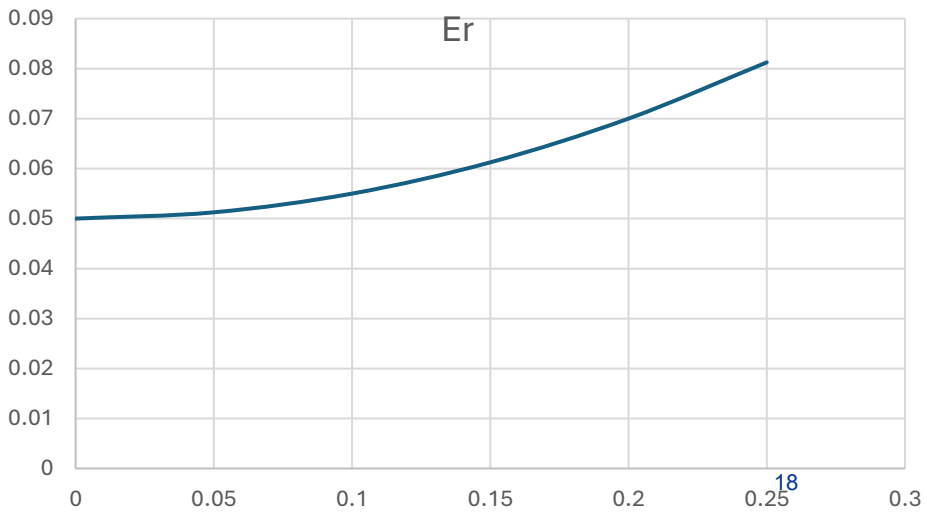
- The indifference curve includes the expected return and variance pairs that provide a given level of utility. To find it, you need to determine a utility level and look for all the combinations of mean and variance that provide that utility. The curve is:

$$Er = U + 0{,}5 \times 3 \times \sigma^2$$

- Example: let's set a utility level of 5%. The risk aversion coefficient is 3. To derive the indifference curve, it is necessary to find the pairs of expected return and variance compatible with the following equation $Er = 0.05 = Er + 0{,}5 \times 3 \times \sigma^2$. We can give some values to the standard deviation and derive the expected return value compatible with utility equal to 5%:

EXAMPLE

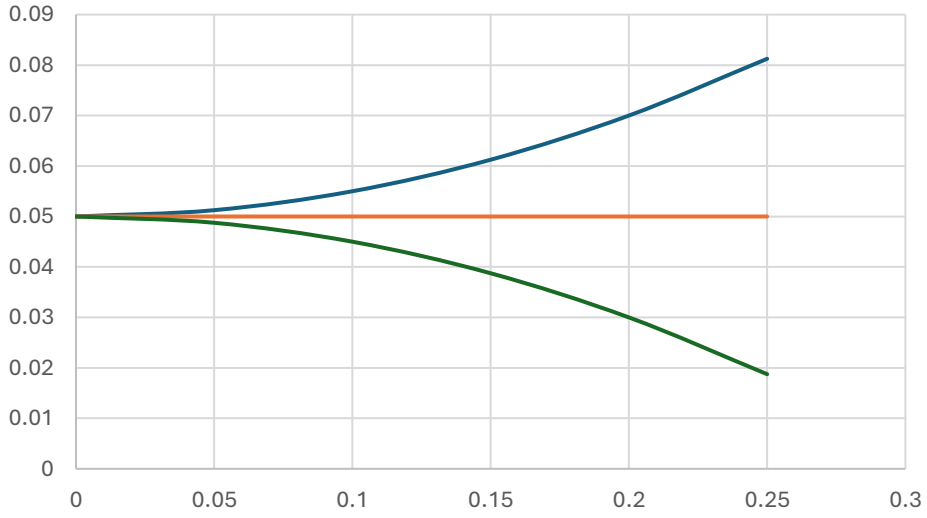
σ	σ^2	E(r), A=3	E(r), A=0	E(r), A=-3
0,00	0,0	0,05		
0,05	0,0025	0,05375		
0,10	0,0100	0,06500		
0,15	0,0225	0,08375		
0,20	0,0400	0,11000		
0,25	0,0625	0,14375		



EXAMPLE

- Derive the indifference curve in the hypothesis A=0 (risk neutrality because the different levels of variance do not affect the level of utility) and A=-3 (propensity to risk because the variance contributes positively to the level of utility)

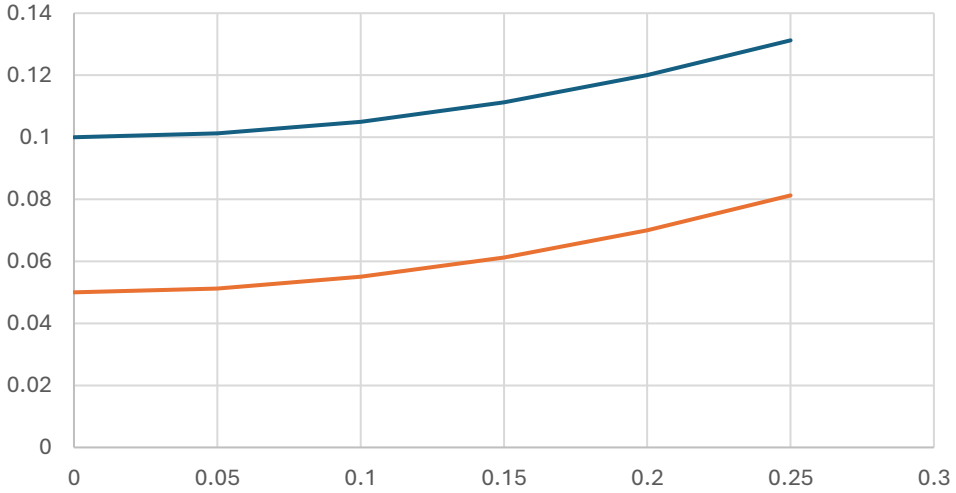
σ	σ^2	E(r), A=3	E(r), A=0	E(r), A=-3
0,00	0,0	0.05	0.05	0.05
0,05	0,0025	0.05375	0.05	0.04625
0,10	0,0100	0.065	0.05	0.035
0,15	0,0225	0.08375	0.05	0.01625
0,20	0,0400	0.11	0.05	-0.01
0,25	0,0625	0.14375	0.05	-0.0438



SHIFTING THE INDIFFERENCE CURVE

- Displacement: We set a utility level of 10%. To derive the indifference curve in the hypothesis A=3, it is necessary to find the expected return and variance pairs compatible with the following equation $0.1=Er-0.5\times3\times\sigma^2$. We can give some values to the standard deviation and derive the expected return value compatible with utility equal to 10%

σ	σ^2	E(r), A=3
0,00	0,0	0,0
0,05	0,0025	0,05375
0,10	0,0100	0,06500
0,15	0,0225	0,08375
0,20	0,0400	0,11000
0,25	0,0625	0,14375



THE SLOPE OF THE INDIFFERENCE CURVE

- The slope of the indifference curve depends on risk aversion: some increase in variance requires a higher expected return increase for a more risk-averse individual so as to leave the same level of total utility

QUESTION

- Question: Using the data in the table above, determine the risk premium for holding a risky security when the risk-free is 5% and the volatility is 5%, and the coefficient A=3. Repeat with A=-3

ANSWER

- Answer: if the risk-free rate is 5%, the utility of the rate is 5%. If the aversion coefficient is equal to 3, we see that for a volatility of 5% the expected return of 5.375% provides a utility level of 5%. It follows that the differential between the expected return of the risky security and the risk-free rate, i.e. 5.375%-5% = 0.375%, is the risk premium. Any expected return level below 5.375% would cause the investor to avoid the risky investment.
If the aversion coefficient is equal to -3, we see that for a volatility of 5% the expected return of 4.625% provides a utility level of 5%. It follows that the differential between the expected return of the risky security and the risk-free rate, i.e. 4.625%-5% = -0.375%, is the risk premium. A risk-loving investor is willing to hold a risky security even when its expected return is below risk-free.

THE RISK PREMIUM AND AVERAGE EXCESS RETURN

- The risk premium translates into average excess returns over the long term. The evidence from the equity market is that the average return is higher than risk-free in the long run, an observation consistent with the market's risk aversion hypothesis

QUESTION

- Question: Does a security have an expected return of 5% and a standard deviation of 18%. The risk-free rate is 2%. What is the maximum value of the risk aversion coefficient A such that the risky security is preferred to the risk-free security?

- Answer: The risky security is preferred to the risk-free one, for a generic value of risk aversion A, in case $0.05-0.5\times A\times(0.18)^2>0.02$. This implies that the maximum value of the risk aversion coefficient is equal to 1.85

INVESTMENT AND TIME HORIZON

- The utility model is static, i.e. it assumes that decisions take place at the beginning of a certain period, and the results are seen at the end of the period. “One period” may therefore be interpretd as “one month”, “one year”, “many years”. Is the preference for stocks over bonds affected by the length of the time horizon?

- With log returns $\overline{r^c_{t+1.t+T}} \equiv \frac{r^c_{t+1.t+T}}{T} = \frac{r^c_{t+1}+r^c_{t+2}+\cdots+r^c_{t+T}}{T}$ the variance is

$$V(\overline{r^c_{t+1.t+T}}) \equiv \frac{V(r^c_{t+1.t+T})}{T^2} = \frac{V(r^c_{t+1})}{T}$$

- Variance drops with time horizon: time diversification. The probability of equity overperforming a fixed risk-free rate increases with the horizon

MEASURING RISK AVERSION

- Banks are supposed to take formal steps to measure their clients' risk aversion under the MiFID regulation (recently extended to consider environmental preferences). This is done through questionnaires that measure several dimensions like knowledge about financial instruments (e.g. derivatives), goals, liabilities and attitudes towards risk

- The results of the questionnaires mix several elements that affect investment behavior: it may signal high tolerance to risk because of elements like knowledge of derivatives, high expected income, no risky liabilities even when the coefficient of risk aversion is large

MEASURING RISK AVERSION

- Psychology has important implications for the estimation of the risk aversion coefficient. Studies show that the adoption of risky behaviors depends on motivational factors, which in turn are influenced by personality, individual history, social events and is expressed differently in various contexts
- Neuro-psychology offers more recent insights associated with the mechanical reasoning in our brain